

SWAPNIL SAHU

DATA ANALYST | Mesa, AZ | (646) 427-4999

sahuswapnil1795@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Data Analyst with 5+ years of experience in transforming complex data into actionable insights using Python, SQL, and BI tools like Power BI and Tableau. Proficient in building scalable ETL pipelines, automating reporting workflows, and optimizing data models in cloud environments such as AWS and Snowflake. Hands-on experience with machine learning, statistical analysis, and Generative AI to support data-driven decision-making. Strong collaborator with a track record of delivering impactful analytics solutions in dynamic, cross-functional teams.

TECHNICAL SKILLS

- Programming & Scripting:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL, PL/SQL, R (Basic)
- Visualization & BI Tools:** Power BI, Tableau, Google Data Studio, Looker, Quicksight, QlikSense
- Databases & Query Engines:** Snowflake, MySQL, PostgreSQL, Oracle, AWS Athena
- ETL & Data Engineering:** dbt, Apache Airflow, Great Expectations, Data Cleaning, ELT Design
- Cloud Platforms:** AWS (S3, Lambda, Athena), Azure, GCP, EMR
- Machine Learning & AI:** XGBoost, SHAP, GPT-4o (OpenAI), Generative AI, K-Means, LangChain (Intro)
- Tools & Environments:** Git, Jupyter Notebook, VS Code, REST APIs, YouTube Data API
- Statistics & Analytics:** A/B Testing, Hypothesis Testing, Linear Regression, Clustering

EXPERIENCE

Data Analyst

COPA Health, Mesa, AZ

Dec 2024 – Present

- Led a Snowflake migration and automated ELT workflows using dbt and Airflow, reducing dashboard latency by 40% and saving \$20K annually.
- Designed interactive Power BI dashboards with advanced SQL and Python, reducing patient intake backlog by 25% and boosting SLA compliance.
- Applied Great Expectations and Snowflake RBAC to enhance data quality from 92% to 98% and ensure audit readiness.
- Deployed an XGBoost model (AUC: 0.87) with SHAP insights to reduce 30-day readmission rates by 12%.
- Implemented a GPT-4o-based reporting assistant to auto-generate executive summaries, saving 8 hours/week and increasing engagement by 20%.
- Mentored junior analysts on Git workflows and agile delivery, improving team throughput by 15%.
- Automated ingestion of semi-structured healthcare data using AWS (S3, Lambda, Athena), reducing prep time by 40%.

Research Analyst

iHARP Lab, University of Maryland, Baltimore, MD

Jul 2024 – Nov 2024

- Analyzed geospatial and climate data using Python (GeoPandas) and ArcGIS to support Arctic environmental modeling.
- Built reproducible data pipelines for remote sensing datasets (NetCDF), improving model efficiency by 30%.
- Produced multivariate climate trend visualizations, contributing to grant proposals and peer-reviewed research.

Data Analyst

Accenture Solutions Pvt. Ltd., Bangalore, India

Feb 2019 – Jul 2022

- Created scalable ETL pipelines and data models in Python and SQL, improving integration across 5+ enterprise systems.
- Automated BI dashboards for sales and churn analytics in Power BI and Tableau, increasing retention by 18%.
- Collaborated with product and business teams to define KPIs and improve reporting accuracy by 25%.
- Investigated data anomalies using root cause analysis and implemented monitoring scripts to proactively flag inconsistencies, enhancing reporting reliability.

Associate Data Analyst

enEXL Technologies & Solutions Pvt. Ltd., Bangalore, India

Jun 2018 – Jan 2019

- Collected, cleaned, and validated raw datasets from internal systems and client sources, improving data quality and reducing reporting errors by 20%.
- Assisted in generating weekly and monthly business reports using Excel and SQL, helping stakeholders track KPIs such as customer retention and operational efficiency.
- Created interactive data visualizations using Tableau to identify trends in customer behavior, contributing to a 15% improvement in campaign targeting accuracy.
- Supported senior analysts in ad hoc analysis and root cause investigation, leading to faster resolution of operational bottlenecks during the quarterly performance cycle.

EDUCATION

- **Master of Science in Information Systems**
University of Maryland, Baltimore County (UMBC), Baltimore, MD
Specialization: Data Science & Artificial Intelligence
Aug 2022 – May 2024 - 3.9/4.0 GPA
- **Bachelor in Computer Science and Engineering**
Central College of Engineering and Management, Raipur, India
Jul 2014 – May 2018

PROJECTS

YouTube Channel Performance Analysis

Python, YouTube Data API

Aug 2024 – Nov 2024

- Built a Python-based data extraction tool using YouTube API to analyze video upload frequency, engagement rates, and content performance.
- Delivered actionable content optimization insights based on video length and viewer interaction trends.

Movie Review Clustering & Pattern Discovery

K-Means Clustering, Python, Scikit-learn

May 2024 – Aug 2024

- Developed an unsupervised ML pipeline using K-Means to group reviews by sentiment and thematic similarity.
- Visualized clusters using Seaborn and Matplotlib to highlight data patterns and insights.

Online Retail Platform Simulation

SQL, PL/SQL

Jan 2024 – Apr 2024

- Designed a relational database simulating e-commerce workflow (catalog, orders, payments).
- Created procedures and triggers in PL/SQL to automate inventory, order, and payment processes.

CERYIFICATION:

- Generative AI for Data-Driven Business Decision-Making – IIM Mumbai
- Financial Services: Climate Change Job Simulation – Oliver Wyman (Forage)
- Data Analytics & Visualization Job Simulation – Accenture (Forage)
- Generative AI Fundamentals – Databricks
- Introduction to Data Analytics – Simplilearn
- Advanced Google Analytics – Google
- Google Analytics for Beginners – Google